LECTURE ACTIVITY 18: CONSERVING HOTEL TOWELS – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) Understand how multiple comparisons can increase the probability of a Type I error.
2) State the null and the alternative hypotheses in terms of "no association" versus "there is an association" as well as in terms of comparing population proportions for five categories of the explanatory variable.
3) Compute the Mean Group Diff statistic from a dataset when comparing multiple proportions.
4) Understand that larger values of the Mean Group Diff statistic suggest stronger evidence against the null hypothesis.
5) Use the 3S strategy with the Mean Group Diff statistic.
6) Describe how to use cards to simulate what outcomes (in terms of difference in sample proportions) are to be expected in repeated random assignments, if there is no association between the two variables.
7) Explain why the simulated null distribution of the Mean Group Diff statistic looks different from other simulated null distributions.
8) Conduct a simulation-based analysis when evaluating claims about multiple proportions
9) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the *p*-value and the study design.

Step 1: Formulate a Research Question

If you have stayed at a hotel recently, you may have noticed how guests are increasingly encouraged to practice conservation habits by not having their towels washed every day— instead hanging them back on the rack to be reused. Are some "encouragement strategies" more effective than others? The research question of interest in this study today is: "Do different socially normed signs produce different rates of towel reuse?"

Step 2: Design a Study and Collect Data

A 2008 study[1] conducted a randomized experiment to investigate how different phrasings on signs placed on bathroom towel racks impacted guests' towel reuse behavior. In particular, the researchers were interested in evaluating how messages that communicated different types of social norms impacted towel

[1] Goldstein, Noah J., et al. "A Room with a Viewpoint: Using Social Norms to Motivate Environmental Conservation in Hotels." *Journal of Consumer Research*, vol. 35, no. 3, 2008, pp. 472–82. JSTOR, https://doi.org/10.1086/586910

reuse. One week prior to a guest staying in the room, rooms at a particular hotel were randomly assigned to receive one of the following five messages on a sign hung on the towel bar in the room:

Message 1: HELP SAVE THE ENVIRONMENT. You can show your respect for nature and help save the environment by reusing your towels during your stay.

- We'll call this the "no social norm," or "**none**" norm

Message 2: JOIN YOUR FELLOW GUESTS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests participated in our new resource savings program by using their towels more than once. You can join your fellow guests in this program to help save the environment by reusing your towels during your stay.

- We'll call this the "**guest** identity norm"

Message 3: JOIN YOUR FELLOW GUESTS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests who stayed in this room (#xxx) participated in our new resource savings program by using their towels more than once. You can join your fellow guests in this program to help save the environment by reusing your towels during your stay.

- We'll call this the "**same room** norm"

Message 4: JOIN YOUR FELLOW CITIZENS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests participated in our new resource savings program by using their towels more than once. You can join your fellow citizens in this program to help save the environment by reusing your towels during your stay.

- We'll call this the "**citizens** norm"

Message 5: JOIN THE MEN AND WOMEN WHO ARE HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 76% of the women and 74% of the men participated in our new resource savings program by using their towels more than once. You can join the other men and women in this program to help save the environment by reusing your towels during your stay

- We'll call this the "**sex** norm".

The study was conducted at a "midsized, midpriced hotel in the Southwest that was part of a national hotel chain" (Goldstein *et al.*, 2008) with 190 rooms. Each room was randomly given one of the five messages above, displayed prominently in the bathroom.

Guests staying at the hotel were not informed they were part of the study; rather, hotel attendants simply recorded, each day, whether or not towel reuse occurred the night before - something the authors refer to as a "potential instance of towel reuse" (Goldstein *et al.*, 2008). The authors examined 1,595 such potential instances of towel reuse. Whenever necessary, we'll treat "reusing towels" as a "success".

Question 1

(a) Identify the explanatory and response variables, and classify them as either categorical or quantitative. Also provide units (if the variable is quantitative) or levels (if the variable is categorical)

- **Explanatory Variable:** Type of Message

  Classification: √ Categorical ◯ Quantitative

  Units or Levels: none, guest, same room, citizens, sex

- **Response Variable:** Reuse

  Classification: √ Categorical ◯ Quantitative

  Units or Levels: reused, did not reuse

(b) Did the study design make use of any pairing of subjects, or are the subjects who receive the different signs independent of each other?

Solution: The observations in the five message groups are independent rather than paired. Each instance of potential towel reuse was associated with one room and one assigned message, and there was no matching of an observation in one message group with an observation in another group.

Spoiler alert - we have two categorical variables! But, in a departure from what we saw in Chapter 5 of the textbook one of these variables is not binary. So, we aren't just comparing across two groups - we're comparing across *multiple*.

But, we still have only one research question: "do different socially normed signs produce different rates of towel reuse?" This leads us to two potential avenues of analysis:

a) Conduct 10 different **pairwise** tests. That is, conduct tests for: (1) none vs. guest, (2) none vs. room, (3) none vs. citizen, (4) none vs. sex, (5) guest vs. room, (6) guest vs. citizen, (7) guest vs. sex, (8) room vs. citizen, (9) room vs. sex, (10) citizen vs. sex.

b) Conduct one hypothesis test that compares all five population proportions at once.

Anyone want to continue down avenue (a)? No? Fair enough! Indeed, choice (b) seems much more appealing - something you should have seen in your reading of Chapter 8 (which we will rely upon heavily in this activity).

Even beyond "laziness" (not wanting to unnecessarily conduct 10 different hypothesis tests to answer a single research question), there is a more statistical/technical reason to prefer approach (b) over approach (a), and it relates to **Type I Errors** and the **significance level**. Recall our working definitions of these concepts:

- Type I Error: Rejecting the null when the null was true

- Significance Level: the value we compare our p-value to when making a determination of whether or not to reject the null hypothesis in favor of the alternative (if the p-value is less than the significance level, reject; otherwise, fail to reject). We usually use the symbol α (the Greek letter "alpha") to denote the significance level.

It turns out that these two notions are intricately linked: **the significance level is the probability of committing a Type I Error**.

⚠ Caution

This is **NOT** saying "the significance level is the probability of rejecting the null hypothesis." Rather; the significance level is the probability of rejecting the null *if the null hypothesis were true*.

So, each time we conduct a test we have an α chance of committing a Type I Error. But, the more tests we conduct, the more we compound our chances of making *at least one Type I Error*. For an analogy, suppose that every time you ski down a certain run, there's a 5% chance you will fall. If you ski down the run 15 times, what's the chance you will fall at least once? What if you ski down the run 100 times?

This relates to a very fundamental problem in statistics, called the **multiple testing** (or **multiple comparisons**) problem. Loosely speaking, the multiple testing problem refers to the fact that: as you increase the number of tests you conduct on the same dataset, you also increase your chances of committing at least one Type I error. This is why it's advisable to not unnecessarily conduct many tests using the same dataset.

Pro Tip

Fun fact: if you use a 5% significance level and conduct 10 tests, there is a 40.13% chance you will commit at least one Type I error. If you conduct 20 tests, the probability jumps to 64.15%.

As such, all things considered, we should use *one* test that compares *all* ~~*four*~~*five* parameters at once. If we fail to reject the null hypothesis, say at the 5% level, we are "done," in that we will conclude that we don't have evidence that the long-run ~~parameters~~proportions differ. By constructing a statistic that compares all sample proportions or sample means at once, we can perform just one test and thus we can keep the probability of a Type I error as small as we want.

Question 2

State the null and alternative hypotheses in words and, if possible, in symbols/math.

Solution: In words:

- H_0 : there is no association between message type and towel reuse
- H_A : there is an association between message type and towel reuse

In symbols:

- $H_0 : \pi_{\text{none}} = \pi_{\text{guest}} = \pi_{\text{same room}} = \pi_{\text{citizens}} = \pi_{\text{sex}}$

where π_{none} is the long-run proportion of instances among the "none" social norm rooms that resulted in successful towel reuse, and similarly for the other π's.

Note: There is no way to write the alternative in symbols.

Step 3: Explore the Data

1) Navigate to the Multiple Proportions applet (you'll need to select "Multiple Proportions" from the list of available applets)

2) In the "**Select Data**" dropdown menu, select "Towels" and click "**Use Table**" (do **not** click "Use Data")

Question 3

You should see the following plot appear:

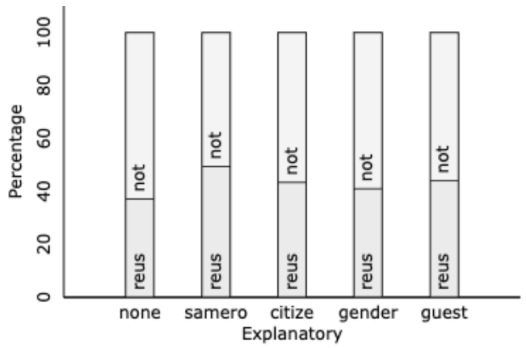


Figure 1: Screenshot from the Multiple Proportions applet

(a) What is the name of this distribution?

- ◯ Population distribution
- ◯ Null distribution
- √ **Sample distribution**

(b) Change the plot to a Mosaic plot (you should remember how to do this from when we last explored this applet). Why does it suffice to only consider the segmented bar graph for this data?

Solution: There are roughly equal counts in each group, so the widths of the rectangles on the mosaic plot will be roughly equal - as such, the "varying widths" aspect of a mosiac plot doesn't give us much more information here.

Looking ahead (from what we saw in the reading, in Chapter 8) we know that the conditional proportions are what will be of great interest to us - as such, we don't need to spend too much time writing down the full two-way table.

Question 4

Fill in the following table with the appropriate information. **Note:** you shouldn't need to do any calculations; all of the required values can be found somewhere in the applet.

Message Type	None	Same Room	Citizen	Sex	Guest	Total
Sample Proportion Reused	0.370	0.493	0.434	0.410	0.441	0.430

Table 1: Summary statistics

Question 5

Looking at the segmented bar graph, as well as the conditional proportions from the table, conjecture on whether you think there will turn out to be a real difference among the five groups with regard to towel reuse.

Solution: The key thing here is to compare the five sample proportions, and see whether we think at least one of them is significantly different from the others. The sample proportions suggest that there may be a real difference among the five groups. For example, my eye is immediately drawn to the fact that the highest towel-reuse proportion was 0.493 (from the same-room message), whereas the lowest towel-reuse proportion was 0.370 (from the no social norm message), which is a seemingly big difference. So, at the very least, I think there is enough to justify an inferential analysis.

Step 4: Draw Inferences Beyond the Data

To investigate, we will see how we can apply the **3S strategy** to this data. Because researchers are attempting to minimize (or at least control) the probability of Type I errors, in cases with multiple groups they first do an overall test. However, this requires computing a single statistic and considering the null distribution of that statistic.

Statistic

Unlike with before, it may not immediately seem like there is a "natural" statistic to consider. As the textbook showed you, one idea is to consider the pairwise differences in conditional proportions - but then there comes the question of how to combine these into a single statistic.

The statistic we'll explore in this activity is the **Mean Group Difference** (aka the Mean Group Diff.) statistic. Here's the procedure we use to calculate this statistic:

1) Calculate all of the pairwise differences in conditional proportions

2) Calculate the absolute value of each of the pairwise differences

3) Calculate the mean of the resulting absolute values

Groups Diff.	Difference in Sample Proportions	Absolute Value	Mean Group Diff.
None – Same Room	0.370 – 0.493 = –0.123	0.123	**0.0554**
None – Citizen	**0.370 – 0.434 = –0.064**	**0.064**	
None – Sex	**0.370 – 0.410 = –0.04**	**0.04**	
None – Guest	**0.370 – 0.441 = –0.071**	**0.071**	
Same Room – Citizen	**0.493 – 0.434 = 0.059**	**0.059**	
Same Room – Sex	**0.493 – 0.410 = 0.083**	**0.083**	
Same Room – Guest	**0.493 – 0.441 = 0.052**	**0.052**	
Citizen – Sex	**0.434 – 0.410 = 0.024**	**0.024**	
Citizen – Guest	**0.434 – 0.441 = –0.007**	**0.007**	
Sex – Guest	**0.410 – 0.441 = –0.031**	**0.031**	

Table 2: Calculating the Mean Group Diff

Question 6

(a) Fill in the second column in Table 2 above.

(b) Fill in the third column in Table 2 above.

(c) Calculate the mean group difference and report it both in the fourth column of Table 2 above as well as in the space below:

- Mean Group Diff: **0.0554**

(d) Check that this matches the value obtained by the applet.

Let's make sure we understand the mean group diff statistic.

Question 7

(a) Can a mean group difference statistic value ever be negative? Explain.

Solution: No. The Mean Group Diff statistic cannot be negative because each difference is changed to its absolute value before the differences are averaged. Absolute values are always 0 or positive.

(b) Can a mean group difference statistic value ever be zero? If so, under what conditions will we observe a value of zero?

Solution: Yes; the mean group difference statistic can be zero, and it will only be zero if each one of the absolute differences is itself zero. That is, the mean group difference statistic will be zero if all conditional proportions are equal to each other.

(c) Which of the following statements is correct?

- √ **Larger values of the Mean Group Diff statistic indicate stronger evidence against the null hypothesis.**
- ◯ Larger values of the Mean Group Diff statistic indicate weaker evidence against the null hypothesis.

Simulate

In Chapter 5, when we were comparing 2 population proportions, we first saw how we can simulate a null hypothesis of no association by shuffling the response variable outcomes across the explanatory variable groups. This models the random assignment process used in the data collection process, and, assuming the null hypothesis is true, "breaks any potential association" between the response and explanatory variables. We modeled this with colored playing cards (or colored poker chips): blue for "successes" and red for "failures."

Question 8

Describe how you would model a simulation of the null hypothesis with red and ~~black~~blue playing cards now that there are five groups instead of two. Make sure to tell how many ~~black~~blue cards ("success") you would need and how many red ones "failure"). Also tell what each color of cards would represent. How many cards would you deal out to each group? What would you calculate after dealing them out?

Solution: Use 1,595 cards, one for each observed instance of potential towel reuse. Use 686 blue cards to represent towel reuse and 909 red cards to represent no towel reuse. Shuffle all the cards and deal them into five groups with the same sample sizes as the original study: 305 cards for none, 306 for same room, 334 for citizen, 310 for sex, and 340 for guest. Then calculate the towel-reuse proportion in each group, find all ten pairwise absolute differences, and calculate their mean to obtain one simulated Mean Group Diff statistic.

Now let's use the Multiple Proportions applet to simulate the null distribution of this Mean Group Diff statistic. This null distribution is again used to measure the strength of evidence provided by the observed statistic against the null hypothesis.

1) Check the box next to **Show Shuffle Options**.

2) Press **Shuffle** to perform one shuffle of the response variable values.

Question 9

You should see the shuffled response variable, the new two-way table for the simulated data, and the value of the simulated statistic (Most Recent Shuffled Mean Group Diff) which is also placed in blue on the graph on the right. Select the Plot radio button to see the shuffled segmented bar graph. How does the distribution across the groups for the shuffled data compare to the original data? Is the simulated Mean Group Diff statistic value closer to zero than the observed value of the Mean Group Diff statistic?

Solution: Answers will vary. In the shuffled data shown in the applet, the group proportions are more similar to one another than in the original data. The shuffled proportions range from about 0.400 to 0.464, whereas the original proportions range from 0.370 to 0.493. The simulated Mean Group Diff is 0.032, which is closer to 0 than the observed value of 0.055.

Update the number of shuffles to 9,999 (for a total of 10,000 repetitions). This repeats the shuffling of the response variable 9,999 more times for a total of 10,000 repetitions.

Question 10

In the applet, you should see a graph of a null distribution. What does each dot represent on the simulated null distribution?

Solution: Each dot represents the Mean Group Diff statistic from one shuffle of the 1,595 reuse and non-reuse outcomes among the five message groups, while keeping the original group sizes fixed. It is one difference-among-groups result that could have occurred by random assignment if the null hypothesis were true.

Question 11

What is the shape of the null distribution? Why is it not a bell-shaped curve centered at zero?

Solution: The null distribution is right-skewed. It is not bell-shaped and centered at 0 because the Mean Group Diff is calculated from absolute differences, so it cannot be negative. Most shuffles produce fairly small differences among the groups, while only a few produce large positive Mean Group Diff values, creating a long right tail.

By making all of the differences in the proportions nonnegative (by taking the absolute value), we cannot get a negative value of the statistic. The result is a right-skewed distribution of statistics that could happen when the null hypothesis of "no association" is true. We have fewer and fewer large values for the statistic, as that would indicate a larger difference among the groups, which shouldn't happen very often when the null hypothesis is true.

Question 12

Is the observed Mean Group Diff statistic out in the tail of this null distribution or is it a fairly typical result? Explain.

Solution: The observed Mean Group Diff of 0.055 is in the right tail of the null distribution. Only a small percentage of the simulated statistics were equal to or larger than 0.055, so the observed result is unlikely if message type has no association with towel reuse.

Question 13

Is the observed Mean Group Diff statistic out in the tail of this null distribution or is it a fairly typical result? Explain.

Solution: The observed Mean Group Diff of 0.055 is in the right tail of the null distribution. Only a small percentage of the simulated statistics were equal to or larger than 0.055, so the observed result is unlikely if message type has no association with towel reuse.

Strength of Evidence

Now that we have a distribution of simulated statistics for the null hypothesis and the value of the observed statistic from our data, we can compute a *p*-value as a measure of the strength of evidence against the null hypothesis. In the past we have used the way we've phrased our alternative hypothesis to determine whether the test is one-sided or two-sided and, thus, how to calculate the *p*-value (one tail or both tails). But in this case, values of the Mean Group Diff statistic that are more extreme than what we observed will always be larger positive values, so we only care about values that are equal to or even larger than the observed value of the statistic.

Question 14

Use the applet to find and report the simulation-based p-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) the simulated null distribution, (2) statistic and (3) *p*-value.

- My simulation-based *p*-value is: **0.0360** **(answers may vary)**

Here's my screenshot:

Solution:

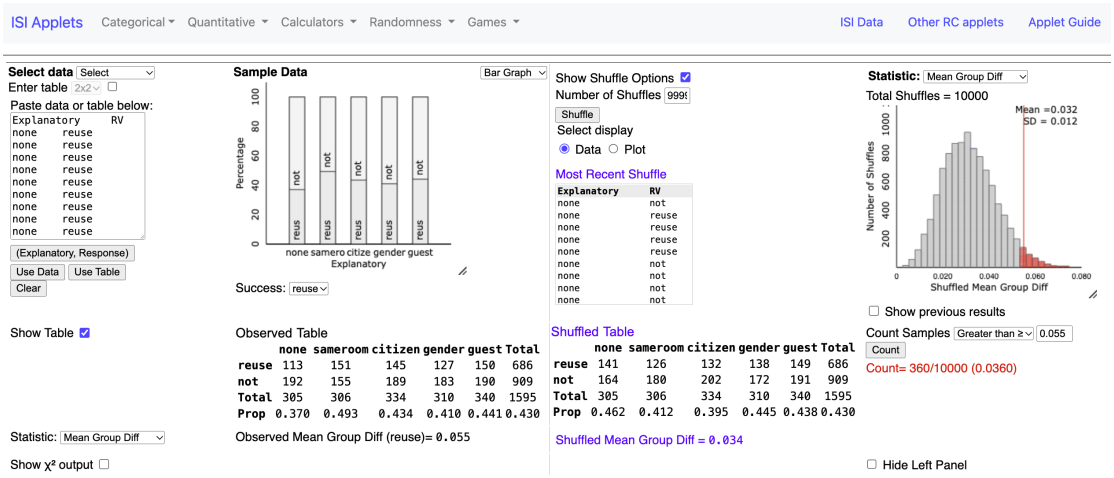


Figure 2: Screenshot from the Multiple Proportions applet

Question 15

Evaluate this *p*-value: How much evidence does the sample data provide against the null hypothesis? Choose one.

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- √ **Strong evidence against the null hypothesis**
- ◯ Very strong evidence against the null hypothesis

Step 5: Formulate Questions

Question 16

Based on your analysis, give an overall conclusion in context. Make sure you refer to the parameter of interest in your conclusion.

Solution: We have strong evidence that at least one population proportion of instances of towel reuse (π) is different across message types, for guests in this hotel or similar guests staying in similar hotels to the one in this study.

Question 17

Generalization: Comment on how the sample was obtained and to which population your conclusion can be generalized.

Solution: The study used 1,595 instances of potential towel reuse from guests at one particular midsized, midpriced hotel in the southwest, rather than a random sample of all hotels or all hotel guests. Therefore, the results should be generalized cautiously to guests and towel-reuse situations similar to those at the hotel in the study, not automatically to every hotel or traveler. Hotels of different levels/stars might have costumers that behave differently than the ones observed in this hotel; for example, we may not feel comfortable generalizing to the *Ritz Carlton* in Los Angeles.

Question 18

Causation: Comment on how the study was designed and whether a cause-and-effect conclusion is warranted from this study.

Solution: Yes, a cause-and-effect conclusion is warranted for settings similar to the study because hotel rooms were randomly assigned to receive one of the five messages. Random assignment tends to balance other variables among the groups, so differences in towel reuse can reasonably be attributed to the message type.

Question 19

Think back to your answer to Question 16. At this point we conclude that at least one population proportion of instances of towel reuse differs from the others. What is a sensible next step? Think as if you were the manager of the hotel, what other information would you want to know?

Solution: A sensible next step is to determine which message types differ from one another and by how much. The hotel manager would want follow-up comparisons, especially comparisons involving the same-room message, which had the highest sample reuse rate, and the no-social-norm message, which had the lowest rate. The manager would also want to consider whether any improvement is large enough to be practically useful.

In the next activity we will learn about the theory-based approach for a test for multiple proportions and then we will try to identify which proportion differs significantly from which others. We can do this by producing confidence intervals for pairwise differences in population proportions.

Step 6: Look Back and Ahead

Question 20

Do you have any concerns about the study design? Any comments on sample size? Are there other limitations that you feel need to be addressed? Are there specific improvements you would suggest to make this study better? Based on what you concluded from this study, can you suggest one or two follow-up studies that could be carried out?

Solution: The randomized design and large sample size are important strengths. However, the observations came from only one hotel, so the findings may depend on that hotel's guests, location, room procedures, or time period. Some guests may also have contributed more than one potential reuse instance, which could make observations from the same guest less independent. A stronger study would include several randomly selected hotels, track individual guests, balance or account for length of stay, and repeat the experiment during different seasons. Follow-up studies could compare the most successful messages directly, test revised wording, and examine whether the effects differ by hotel type or guest population.

Segue: Sigma Notation

From here on out, the course is going to start getting a little more mathematical; in particular, we'll start relying more and more on mathematical formulas. To ease us into this, I wanted to take a bit of time to introduce one mathematical notation we'll see a lot: **sigma notation**. Sigma notation is a handy shorthand notation we use to write sums.

First: recall that we use n as a placeholder for "sample size" when we don't have a specific numerical value to work with. Analogously, we often use the symbol x_i as a place holder **for a piece of data/value** when we don't have a specific numerical value to work with. For example, we can denote an arbitrary sample by

$$x_1, x_2, \cdots, x_n$$

Again, this is just a useful notation to have when we want to talk about "a sample," without having a concrete sample to work with.

Given a sample $x_1, x_2, \cdots, x_n$, one way to write the sum of the sample values is as

$$x_1 + x_2 + \cdots + x_n$$

But this is a little clunky, and takes a bit of time. Sigma notation says we can succinctly write

$$\sum_{i=1}^{n} x_i$$

to mean $x_1 + x_2 + \cdots + x_n$. Why the name "Sigma notation?" Because the symbol Σ is a capital Greek letter Sigma (we've seen the lowercase version of this letter before - σ, the population variance!)

Pro Tip

Anytime you see the symbol $\sum_{i=1}^{n} x_i$, just think: "this is a shorthand way of telling me to add up the elements in my sample."

Question 21

Write the formula for calculating $\bar{x}$, the sample mean, in sigma notation.

Solution:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

If you're curious - and this is **NOT** a formula you need to know for this course - the formula for calculating s, the sample standard deviation, is

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$